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1 - OVERVIEW

See Illustration 1-1 below for a picture of the front of the RFI. The frequency of operation of the 1.5T RFI (GE Part # 2255388) is 63.86 MHz +/- 500 KHz. The frequency of operation of the 1.0T RFI (GE Part # 2194013) is 42.60 MHz +/- 500 KHz. The Radio Frequency Interface (RFI) has 3 main functions. The first is to provide a way to adjust the gain of the head and body RF power output. The second is to split the RF signal from the exciter and provide a separate RF drive signal for each amplifier so that the RF output from both will be equal in magnitude and phase. Phase and gain adjustments are provided on the RFI so as to get equal and in-phase RF power output from each RF amplifier and, as a result, maximum power output from the RFI Combiner. The RFI Combiner is located inside the RFI and combines (sums) the in-phase output of the 2 RF amplifiers to provide the proper RF output power (Gain) needed for Body or Head mode.

Note

In the RF/PDU and SRF Cabinets the RF signal from the exciter is connected via a rear interface bracket to BNC connector J14 on the front of the RFI Module. The Envelope Feedback circuitry inside the System Support Module (SSM) is not utilized. There is, therefore, no connection to J105 or J103 of the SSM.

The third function is to act as a communications interface between the RF amplifiers and the SSM. Status and failure information from the amplifiers are received and interpreted in the RFI and then transferred to the System Support Module (SSM) and ultimately to the host computer.

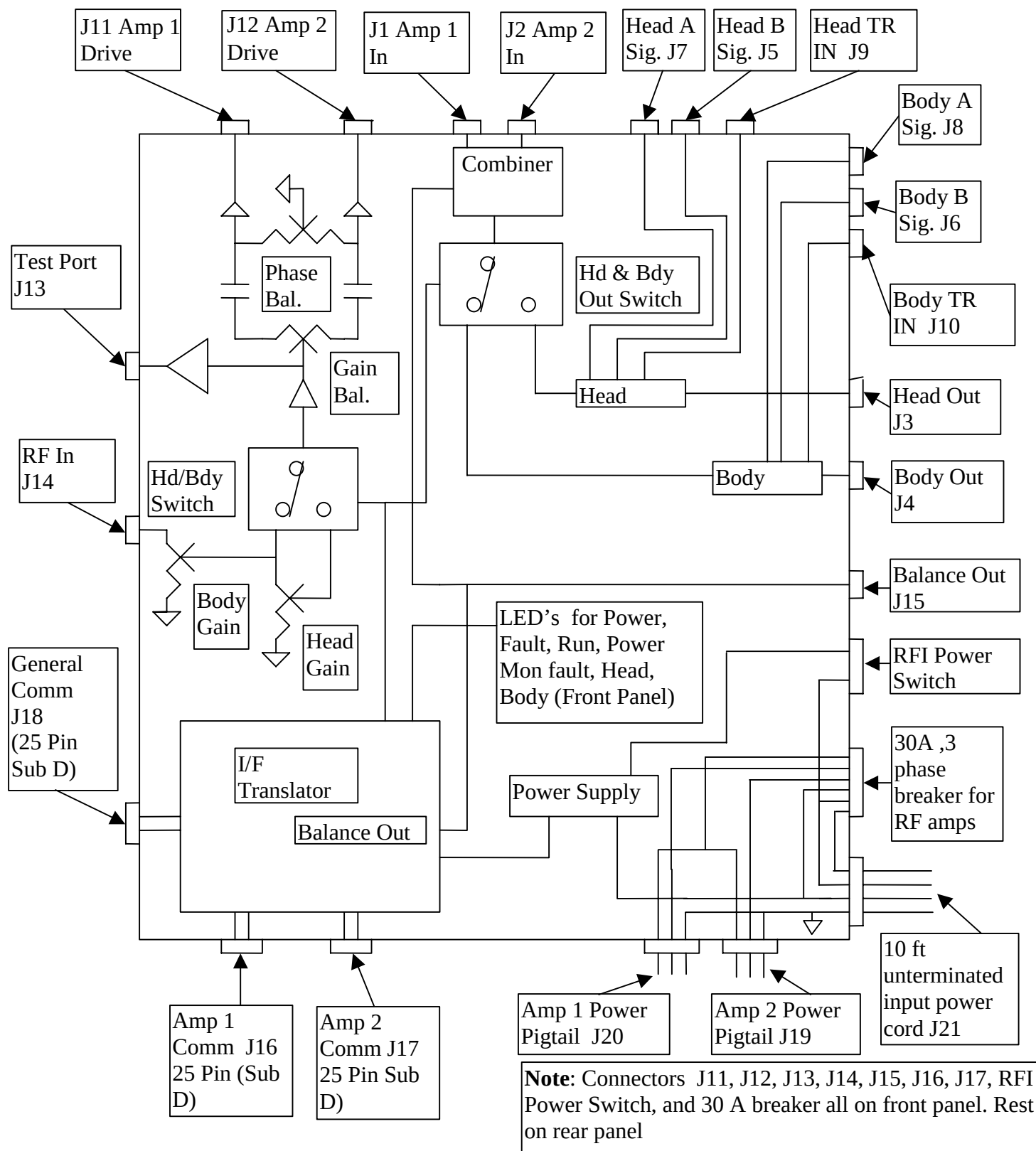
The RFI consists of four modules; Input conditioning module, Output conditioning module, Interface Translator module and a commercially available power supply. Illustration 1-2 is a functional block diagram of the RFI showing the three functional blocks: IF Translation, Input Conditioning, and Output Conditioning.

This document is concerned with the 1.0T RFI (GE Part # 2194013) and the 1.5T RFI (GE Part # 2255388). While the front panels of the 1.0T RFI and 1.5T RFI look identical, the J1, J2 and J4 RF ports on the rear of the 1.0T RFI are N connectors while the same ports on the rear of the 1.5T RFI are HN connectors. For this and other reasons the 1.0T RFI and 1.5T RFI are, obviously, NOT interchangeable. The GE Part #'s listed on the manufacturers label, located in the upper left rear of the RFI, can also be used to differentiate between the 2 RFI types.



FRONT VIEW OF RFI (FRONT IS IDENTICAL BETWEEN 1.0T AND 1.5T MODELS)
ILLUSTRATION 1-1

1 – OVERVIEW (Continued)



RFI FUNCTIONAL BLOCK DIAGRAM
ILLUSTRATION 1-2

2 – RFI FRONT PANEL

The components present on the RFI front panel are described in this section.

2-1 Front Panel RFI Switches And Fuses

Refer to Illustration 2-1.



RFI BREAKER, SWITCH, AND FUSES
ILLUSTRATION 2-1

2-1-1 Amplifier Power Circuit Breaker

This is a 208VAC, 3-phase, 30 Amp, circuit breaker that provides a means for removing single-phase line power from both RF amplifiers without removing line power from the RFI.

2-1-2 RFI Power Switch

This is a toggle switch that provides a means for removing line power from the internal DC power supply inside the RFI without removing line power from the RF amplifiers.

2-1-3 RFI Front Panel Line Fuses

Two Fast Blow line fuses rated at 250VAC, 0.4 Amps each are accessible from the front panel of the RFI. The failure of either fuse will result in the fan failing to rotate and the loss of +/- 15VDC and +5VDC from the internal RFI DC power supply.

2-2 RFI LEDs

Refer to Illustration 2-2.



RFI LEDs, CONNECTORS, AND ADJUSTMENT POTENTIOMETERS
ILLUSTRATION 2-2

2-2-1 PWR LED

This LED is illuminated whenever the RFI is powered on. This LED will not be illuminated when the RFI Power switch is in the down (OFF) position.

2-2-2 FAULT LED

This LED is illuminated whenever the RFI detects a fault condition. The fault condition could originate from the RFI or one of the RF amplifiers. This LED will not illuminate from a fault condition originating in the SSM.

2-2-3 RUN LED

This LED is illuminated whenever the system is in Ready mode or when the system is pulsing.

2-2-4 PMON LED

This LED is illuminated whenever a power monitor fault is detected by the SSM and reported to the RFI.

2-2-5 BODY LED

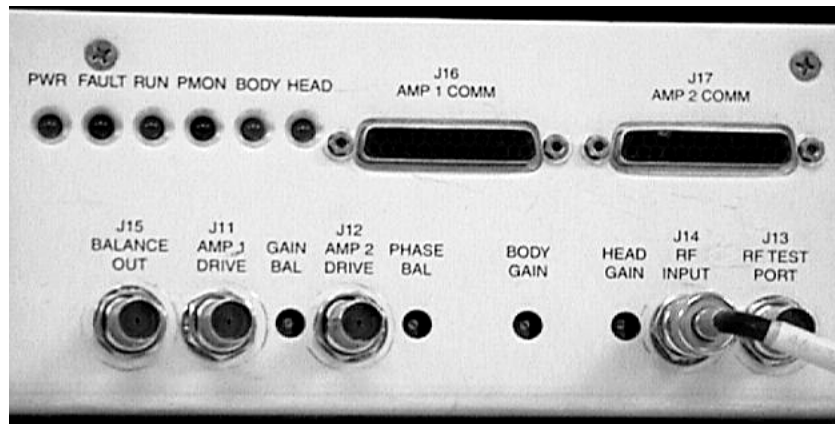
This LED is illuminated while the RFI is in body mode.

2-2-6 HEAD LED

This LED is illuminated while the RFI is in head mode.

2-3 Front Panel RFI Connectors And Adjustment Controls

Refer to Illustration 2-3.



RFI FRONT PANEL CONNECTORS
ILLUSTRATION 2-3

2-3-1 J11 AMP 1 DRIVE Port

Gain and phase balanced RF is output from this female BNC port to the input of RF amplifier 1. The output is typically on the order of -10dBm to -9dBm for head and -1dBm to 0dBm for body.

2-3-2 J12 AMP 2 DRIVE Port

Gain and phase balanced RF is output from this female BNC port to the input of RF amplifier 2. The output is typically on the order of -10dBm to -9dBm for head and -1dBm to 0dBm for body.

2-3-3 J13 RF TEST PORT

If the RFI is placed into TEST mode (signified when the HEAD and BODY LEDs are both extinguished but the PWR LED is illuminated) and 0dBm (1 mW) is input to J14 RF INPUT port then 26.02dBm ($400\text{ mW} \pm 10\%$) will be output from the female BNC J13 RF TEST PORT. This port is used for certain system tests like High Speed Stability (HSS).

2-3-4 J14 RF INPUT Port

RF signal from the Exciter in the System Cabinet is routed into the RFI from this female BNC port. The RF signal input is typically from 0dBm to 4dBm for head or body scanning. The RF Power Output Calibration documents each provide a section that describes how to verify that the correct level of RF voltage is reaching this port from the (U)CERD.

2-3-5 J15 BALANCE OUT Port

This female BNC port provides a reference voltage that will typically range from $8 - 11\text{ VDC}$ and is used to determine whether the RF amplifiers are properly balanced or not. This voltage reacts slowly to change and it is usually necessary to wait about 1 minute for it to stabilize before taking a reading.

2-3-6 J16 AMP 1 COMM Port

This female 25-pin Sub-D connector provides the status, control, and failure communication link between the RFI and RF amplifier 1.

2-3-7 J17 AMP 2 COMM Port

This female 25-pin Sub-D connector provides the status, control, and failure communication link between the RFI and RF amplifier 2.

2-3-8 GAIN BAL Potentiometer

This 30-turn potentiometer provides a means for balancing the gain difference between the RF amplifiers when performing the RF Output Power Calibration. Detents are provided at either extreme of the pot turning range.

2-3-9 PHASE BAL Potentiometer

This 30-turn potentiometer provides a means for balancing the phase difference between the RF amplifiers when performing the RF Output Power Calibration. Detents are provided at either extreme of the pot turning range.

2-3-10 BODY GAIN Potentiometer

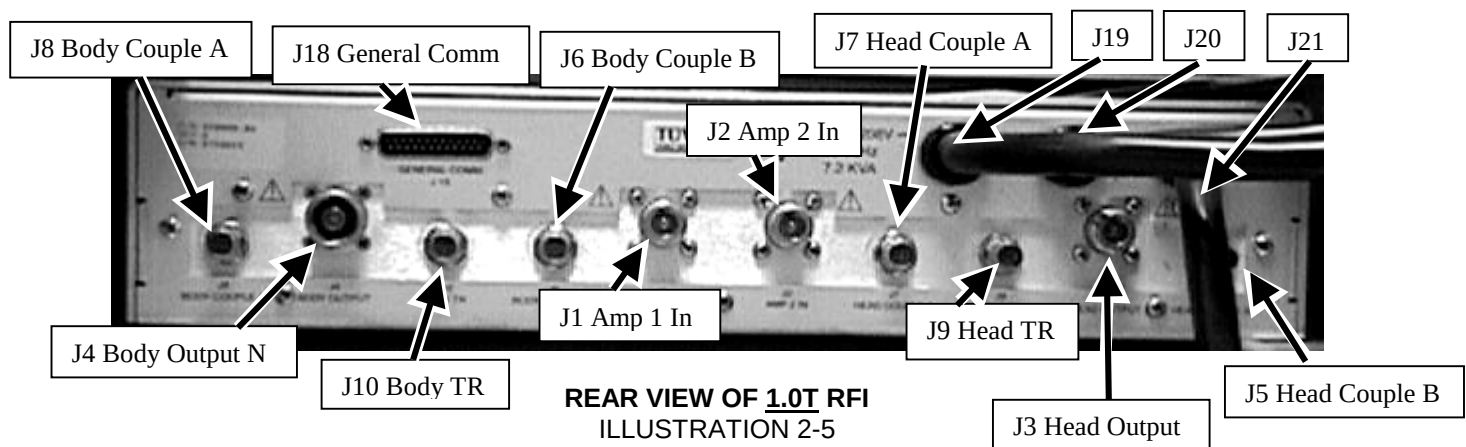
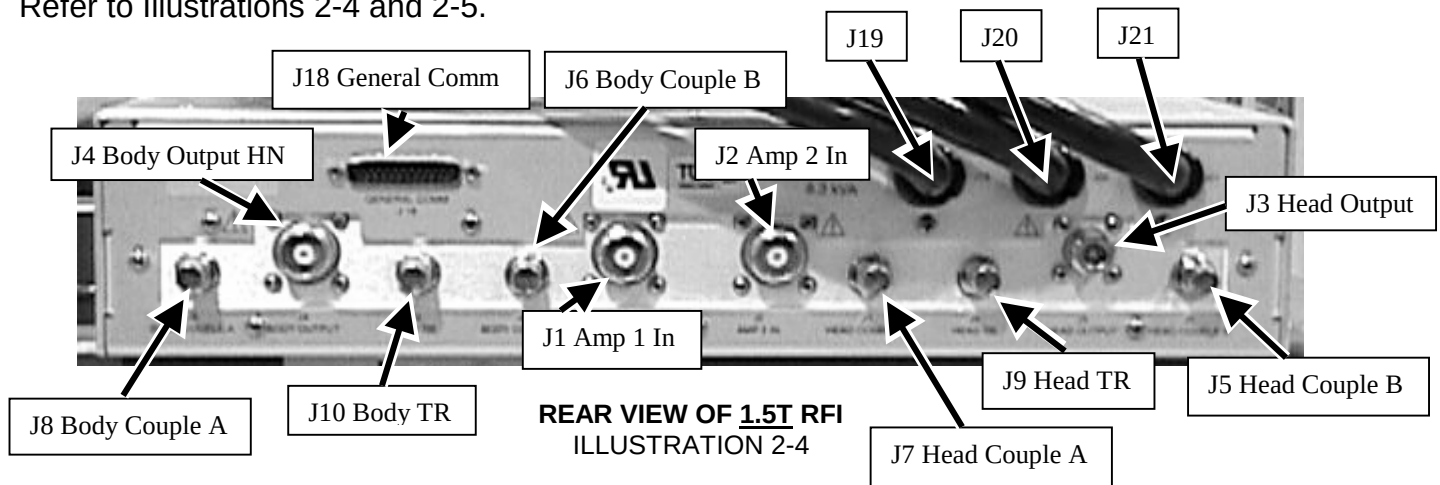
This 30-turn potentiometer provides a means for adjusting the body RF output power from the RF amplifiers. This is adjusted when performing the RF Output Power Calibration for body. Detents are provided at either extreme of the pot turning range.

2-3-11 HEAD GAIN Potentiometer

This 30-turn potentiometer provides a means for adjusting the head RF output power from the RF amplifiers. This is adjusted when performing the RF Output Power Calibration for head. Detents are provided at either extreme of the pot turning range.

2-4 Rear Panel RFI Connectors

Refer to Illustrations 2-4 and 2-5.



2-4-1 J1 AMP 1 IN Port

RF output from RF amplifier 1 (top amplifier in the cabinet) is routed into the RFI combiner module from this port. On a **1.5T RFI** (GE Part # 2255388) this port will be female HN connector. On a **1.0T RFI** (GE Part # 2194013) this port will be a female N connector.

2-4-2 J2 AMP 2 IN Port

RF output from RF amplifier 2 (bottom amplifier in the cabinet) is routed into the RFI combiner module from this port. On a **1.5T RFI** (GE Part # 2255388) this port will be female HN connector. On a **1.0T RFI** (GE Part # 2194013) this port will be a female N connector.

2-4-3 J3 HEAD OUTPUT Port

Head RF output power from the RFI Combiner and TR bias is output from this female N connector. Isolation between the head and body ports should be greater than 40dB.

2-4-4 J4 BODY OUTPUT Port

Body RF output power from the RFI Combiner and body TR bias is output from this port. This port is a male N connector on a 1.0T RFI and a female HN connector on a 1.5T RFI. Isolation between the head and body ports should be greater than 40dB.

2-4-5 J5 HEAD COUPLE B Port

This female BNC connector provides a sample of the RF output that is routed, through RF cable, to the head B sample port on Power Monitor B in the SSM. The directivity of the sampled signal is greater than 15dB. The amount of sampled signal available at this port depends on whether the RFI is a 1.5T (part # 2255388) or 1.0T (part #2194013) unit. See Table 2-1 for a summary of the sampled output levels.

TABLE 2-1
1.5T AND 1.0T RFI HEAD COUPLE SAMPLED SIGNAL OUTPUT LEVELS

RFI TYPE	RFI PART NUMBER	HEAD COUPLE SAMPLED SIGNAL OUTPUT
1.5T	2255388	-40dB (+/- 0.5dB)
1.0T	2194013	-35dB (+/- 0.5dB)

2-4-6 J6 BODY COUPLE B Port

This female BNC connector provides a sample of the RF output that is routed, through RF cable, to the body B sample port on Power Monitor B in the SSM. The directivity of the sampled signal is greater than 15dB. The amount of sampled signal available at this port depends on whether the RFI is a 1.5T (part # 2255388) or 1.0T (part #2194013) unit. See Table 2-2 for a summary of the sampled output levels.

TABLE 2-2
1.5T AND 1.0T RFI BODY COUPLE SAMPLED SIGNAL OUTPUT LEVELS

RFI TYPE	RFI PART NUMBER	BODY COUPLE SAMPLED SIGNAL OUTPUT
1.5T	2255388	-50dB (+/- 0.5dB)
1.0T	2194013	-46dB (+/- 0.5dB)

2-4-7 J7 HEAD COUPLE A Port

This female BNC connector provides a sample of the RF output that is routed, through RF cable, to the head A sample port on Power Monitor A in the SSM. The directivity of the sampled signal is greater than 15dB. Since this port and the J5 Head Couple B Port are redundant ports, the amount of sampled signal available at this port should, ideally, be the same as that available at the J5 Head Couple B Port. See Table 2-1 for a summary of the sampled output levels.

2-4-8 J8 BODY COUPLE A Port

This female BNC connector provides a sample of the RF output that is routed, through RF cable, to the body A sample port on Power Monitor A in the SSM. The directivity of the sampled signal is greater than 15dB. Since this port and the J6 Body Couple B Port are redundant ports, the amount of sampled signal available at this port should, ideally, be the same as that available at the J6 Body Couple B Port. See Table 2-2 for a summary of the sampled output levels.

2-4-9 J9 HEAD TR Port

Head TR bias from the SSM is introduced into the RFI and, ultimately, onto the J3 HEAD OUTPUT port from this female BNC connection.

2-4-10 J10 BODY TR Port

Body TR bias from the SSM is introduced into the RFI and, ultimately, onto the J4 BODY OUTPUT port from this female BNC connection.

2-4-11 J18 GENERAL COMM Port

Status, control and fault information from the RFI and both RF amplifiers is transferred through this 25-pin male Sub-D connection to the SSM.

2-4-12 J19 RF Amplifier 1 Power Cable

Single-phase, 208VAC line power is supplied to RF amplifier 1 from this 3-conductor cable.

2-4-13 J20 RF Amplifier 2 Power Cable

Single-phase, 208VAC line power is supplied to RF amplifier 2 from this 3-conductor cable.

2-4-14 J21 RFI AC Input Power Cable

3-phase, 208VAC line power is supplied on this 4-conductor cable to the RFI.

In the **1.0T RF/PDU Cabinet** (GE Part # 2174501) and **1.5T RF/PDU Cabinet** (GE Part # 2224560) this power comes from the NB AMP connector on the rear of the SSM.

In the **1.0T SRF Cabinet** (GE Part # 2174500-2) and the **1.5T SRF Cabinet** (GE Part # 2174500) this power comes from the CB1 circuit breaker located in the rear of the SRF Cabinet.

3 - INTERFACE TRANSLATOR MODULE AND FIRMWARE

The Interface Translator is the communication link between the SSM and the RF Amplifiers. The GENERAL COMM interface between the SSM and the RFI includes RS-232, UNBLANK, and power monitor fault signals. The power monitor fault signals, MON1, MON2, and HVRLY, are generated by the SSM. The SSM is also the source of UNBLANK.

The interface to the RF Power Amplifiers (AMP 1 COMM and AMP 2 COMM) contains amplifier control signals such as MAINPWR1 (amplifier enable signal), RESET (for resetting faults), and UNBLANK as well as 6 bits of read-only amplifier status information (STATUS 0 - STATUS 5). MAINPWR1 and UNBLANK signals are outputs of the ISPLSI2032 EPLD. The RESET signal originates in the MC68HC811E2FN microprocessor. Amplifier status information (STATUS 0 - STATUS 5) is read by the microprocessor and made available to the GENERAL COMM interface via RS-232 for error reporting. Amplifier status bits (STATUS 0, STATUS 1, STATUS 5) are also used by the EPLD as a condition for unblanking the amplifiers and asserting the MAINPWR1 enable signal.

The Translator can be functionally divided into two sections: the interface with the RS-232 port and the interface with the amplifier port. There is only a small overlap between these two functions. The MC68HC811E2FN (and the associated firmware) handles all of the RS-232 interface functions and a portion of the amplifier interface while the ISPLSI2032 (EPLD) handles only amplifier related functions. The interface between the microprocessor and the EPLD is a small number of digital signals that appear in Table 3-1 below. Also listed in Table 3-1 are signals from the microprocessor that directly control the RF Amplifiers via the AMP 1 COMM and AMP 2 COMM ports.

3 - INTERFACE TRANSLATOR MODULE AND FIRMWARE (continued)

TABLE 3-1
CONTROL SIGNALS FROM MICROPROCESSOR

Signal Name	Signal Direction	Description
RESET	from micro to output buffer (AMP 1 COMM & AMP 2 COMM)	Set whenever the RFI is in the "off" state. Cleared otherwise
/MAINPWR2 (AC1)	from micro to output buffer (AMP 1 COMM & AMP 2 COMM)	always the opposite of OPERATE command.
OPERATE	from micro to EPLD	Set whenever the RFI is in the "operate" state; cleared while in "off" or "standby"
BODYMODE	from micro to EPLD	Set in Body or Test modes, cleared in Head mode.
TESTMODE	from micro to EPLD	Set in Test mode, cleared in Head or Body mode.
NOT-LOCAL-FAULT	from micro to EPLD	Set whenever the micro 'sees' no faults.

The EPLD controls the RF Amplifiers via the MAINPWR1 and UNBLANK signals. As shown in Table 3-2 below, the value of these signals is dependent on signals from the microprocessor and signals (MON1, MON2, and HVRLY) from the GENERAL COMM port.

TABLE 3-2
Control Signals from EPLD

Signal Name	Signal Direction	Description
/MAINPWR1 (AC2)	from EPLD to output buffer (AMP 1 COMM & AMP 2 COMM)	low if OPERATE signal (from micro) is high and no power monitor faults have been detected on the GENERAL COMM port (MON1=0, MON2=0, HVRLY=0).
/UNBLANK	from EPLD to output buffer (AMP 1 COMM & AMP 2 COMM)	unblanking occurs if /MAINPWR1 is low, the RF amplifier is not reporting any faults (AMP1 S5=0, AMP2 S5=0), both RF amplifier are in operate mode (AMP1 S0=1, AMP2 S0=1, AMP1 S1=0, AMP2 S1=0), and NOT-LOCAL-FAULT=1

3-1 RS-232 functions

The micro, U9, parses all of the RS-232 commands and converts them into corresponding digital signals. In addition, the micro enables a periodic timer interrupt (running at 1.00kHz) for the purpose of assessing most of the fault conditions. Specifically, the functions handled by the micro, divided into 2 categories, are:

Note

The EPLD schematic is needed to follow most of these signals off the board.

Outputs:

MW3001 00:	Results in setting the “off” bit (Port B bit 3) and clearing the “operate” bit (Port B bit 7).
MW3001 01	Results in clearing both the “off” bit and “operate” bit.
MW3001 03	Results in clearing the “off” bit and setting the “operate” bit.
MW3002 20	Results in setting the “test mode” bit (Port B bit 5).
MW3002 40	Results in clearing the “test mode” bit and setting the “head mode” bit (Port B bit 4).
MW3002 60	Results in clearing both the “test mode” and “head mode” bits.
MW3003 XX	Results in assigning the value XX to the frequency code. This will overwrite the default value of 0x34.
Not Local fault:	Updated by the processor asynchronously by the main timer ‘tick’ interrupt routine every 1.00ms. This bit (Port B bit 1) is cleared whenever the processor is aware of the presence of any fault (other than “mode fault”). It is otherwise always set. All faults tested by this condition are described below. Note: There is only one fault condition (the mode fault) which is not sampled by the main timer interrupt.

3-1 RS-232 Functions (continued)

Status related:

MR3011 Results in reading the “status byte” which is fully defined in the specifications

MR3014 Results in reading and transferring the “fault” byte. A complete list of fault codes decoded by the translator are listed in Table 3-2.

TABLE 3-2
FAULT CODES

Fault Code (In Hex)	Brief Description	Detailed Description
\$10	mode fault	a command was received to change the mode of the RFI while in “operate”. The command was acted upon only after blanking the system and returning to “off”.
\$20	dump resistor over temp	A thermistor is mounted near the power dump resistor. As this resistor heats, the voltage on Port E bit 7 should go down. The signal is sampled inside the main timer interrupt (every 1.00 ms). A software filter accumulates the average of the last 16 A/D samples. The fault indicates that this average fell below the fault threshold (2.00V at the time of this writing).
\$21	mismatch	The mismatch voltage is presented to the processor at Port E bit 6; an A/D converter input. The signal is sampled inside the main timer interrupt (every 1.00 ms) and is acted upon immediately without software filtering. The fault indicates that the signal exceeded the fault threshold (3.75V at the time of this writing).
\$22	disconnected from amp 1	Port C bit 7 was found high.
\$23	disconnected from amp 2	Port C bit 6 was found high
\$30	moon 1 fault	Port A bit 0 was found high
\$31	moon 2 fault	Port A bit 1 was found high
\$32	hv relay fault	Port D bit 3 was found high
\$40-\$5F	reserved for amp 1	Status bit 5 (seen at Port C bit 5) indicates the fault status of amp 1. Bits 0-4 indicate the actual fault information.
\$60-7F	reserved for amp 2	Status bit 5 (seen at Port E bit 5) indicates the fault status of amp 2. Bits 0-4 indicate the actual fault information.

4 - INPUT CONDITIONING MODULE

The Input Conditioning Module is the low-power RF section of the RF Interface. Its purpose is to perform the primary functions listed below:

1. Head/Body Drive Level Setting
2. Blanking and Mode Selection
3. Drive and Test Signal Amplification
4. Amplifier Drive Level and Phase Balancing

In addition to the primary functions listed above, the Input Conditioning Module also serves as an interface between the Processor Module and all other RFI circuitry. This interface includes four digital control signals: HEAD, BODY, TEST, and UNBLANK. Test points are installed on the Input Conditioning Board for each of these four control signals. The +5VDC supply for the CPU is routed through the Input Conditioning Board to the Processor Module. The BALANCE signal, which indicates the power dissipated in the dump resistor of the Output Conditioning Module is scaled, clamped, and peak detected on the Input Conditioning Board before being passed to the Processor Module on pins J25-10.

4-1 Head/Body Drive Level Setting

The RF INPUT (J14) is AC coupled by capacitor C1. Head and body drive levels are set by way of two variable resistive attenuators. R2 and the potentiometer (R7) form a voltage divider that is used to set the body drive level. This divider is adjustable between 0dB and 16dB of attenuation so that a 10dB range of input power levels is easily accommodated. The body mode drive signal passes from the wiper of potentiometer R7 to pin #5 of the analog multiplexer (U1). D1 and D2 protect the module from overvoltage conditions on the input. The head mode drive signal (pin #4 of U1) is taken from the wiper of potentiometer R12 which, together with R11 and R13, forms a divider and attenuates the body mode drive signal by an additional 7dB to 13dB. As a result of this arrangement, the head drive level is directly affected by the body drive level adjustment. Therefore, the body drive level must always be set before the head drive level.

Note

The Body and Head Gain pots are in series with each other. The Body Gain pot is first in series. The Head Gain pot is factory set for a 10 dB delta (below) in relationship to the Body Gain. The result is that most sites will ONLY require adjustment of the Body Gain pot (the Head Gain will follow accordingly).

4-2 Blanking and Mode Selection

The HEAD, BODY, and TEST signals are all active high and only one is high at any point in time. The status of these signals determines which mode the RFI is operating in. The HEAD and BODY LEDs, which are part of the RFI Processor Board, are visible from the front of the RFI and indicate which mode the RFI is in. If both LEDs are off, the RFI is in test mode.

Drive signal blanking is accomplished by the analog multiplexer (U1) that is also used to select between head and body drive levels. In order to achieve better than 40dB of blanking, a 74HC4052 dual analog multiplexer is used with the output of the first switch fed back to the inputs of the second switch. The multiplexers in the 74HC4052 are 1 of 4 switches. Only two of the inputs to each switch are used. All unused inputs are grounded.

The channel select inputs of the 74HC4052 are driven by the UNBLANK and HEAD control signals which pass from the Processor Board to the Input Conditioning Board on pins J25-9 and J25-6, respectively. When the HEAD signal is high AND the UNBLANK signal is high, the head mode drive signal (on pin #4 of U1) is connected to the output of U1 (pin #13). When the HEAD signal is low AND the UNBLANK signal is high, the body mode drive signal (on pin #5 of U1) is connected to the output of U1 (pin #13). For all other conditions the output of U1 is connected to the unused (grounded) inputs.

The TEST control signal passes from the Processor Board to the Input Conditioning Board on pin J25-7. For test mode, the TEST signal is high. Q1 turns on and pulls the gate of Q2 low. Q2 turns on and supplies 15V to the supply pin of the MHW591 (U3), thus allowing the pulsed drive signal to be amplified and emerge at the RF TEST PORT (J13).

4-2 Blanking and Mode Selection (continued)

Head/Body Relay Drivers

The BODY control signal passes from the Processor Board to the Input Conditioning Board on pin J25-5. This signal and the HEAD signal are used to drive the head and body relay drivers. When the HEAD signal is high, Q4 turns on and pulls down the base of Q3 which turns on Q3. The coil of the head relay is connected from the collector of Q3 to the -15V supply. Therefore, when Q3 is turned on, the voltage across the coil is approximately 20V and the head relay is energized. The body relay driver functions in an identical way.

4-3 Drive and Test Signal Amplification

From the output of U1, the drive signal goes to the input of a CLC449 op-amp gain stage (U4). This stage amplifies the drive signal by approximately 12dB. The gain is incorporated to compensate for losses due to the splitting of the drive signal and output matching, so that the Head/Body Drive Level Setting and the Amplifier Drive Level Balancing adjustments each have an effective top end of +3dB.

The input drive to the MHW591 is taken from the wiper of potentiometer R4 which, together with R6 and R16, forms a divider to attenuate the drive signal by 9dB to 18dB. This allows the output of the test amplifier to be set at precisely 400mW (26.02dBm) with a nominal 1mW (0dBm) signal at the RF INPUT (J14).

Note

In TEST mode the amp drive is active at the body power level, but the unblank signal is removed.

4-4 Amplifier Drive Level and Phase Balancing

The AMP 1 DRIVE and AMP 2 DRIVE output levels are adjustable to allow an amplitude balance adjustment of $\pm 3\text{dB}$ (nominal) and a phase balance adjustment of $\pm 40^\circ$ (nominal). These adjustments are made available so that the drive levels can be trimmed to compensate for inequalities in the amplitude and phase response of the RFI splitter circuitry. Both the phase and gain adjustments are differential in nature; both amplifier drive outputs (J11 and J12) are affected by each adjustment. Amplifier imbalance is gauged by comparing the voltage at the J15 BALANCE OUT port when the system is not scanning and is "idle" with the voltage at the J15 BALANCE OUT port when the system is scanning and is "active". Ideally, the "idle" and "active" voltages should be identical if the amplifiers are properly balanced. Differences up to 0.5 Volts are considered inconsequential. Differences over 0.5 Volts indicate that the amplifiers are not sufficiently balanced. The level of the measured J15 voltage is inversely proportional to the power dissipated in the dump resistor. That is, the highest voltage obtainable (usually in the 9 – 11 Volt range) indicates that very little power is being dissipated in the dump resistor and, therefore, the amplifiers are properly balanced. The "highest obtainable voltage" varies from RFI to RFI due to site environmental and unit component differences. The "highest obtainable voltage" and the minimum amount of power dissipated in the dump resistor are achieved by varying the PHASE BALANCE and GAIN BALANCE potentiometers on the front of the RFI until the voltage on the J15 BALANCE OUT port is maximized. The GAIN BALANCE signal is taken from the wiper of R21 which allows a $\pm 3\text{dB}$ (nominal) variation in drive level, while the PHASE BALANCE signal is taken from the wiper of R17 and allows a $\pm 40^\circ$ (nominal) variation in the phase of the drive signal. These signals pass through a CLC449 differential amplifier (U6 and U7). 47.5 ohm resistors (R27 and R28) are placed in series with the output of each amplifier to create a near 50 ohm source impedance. The AMP 1 DRIVE (J11) and AMP 2 DRIVE (J12) outputs are AC coupled through C18 and C24.

5 - OUTPUT CONDITIONING MODULE

The Output Conditioning Module is the high-power section of the RF Interface. Its purpose is to perform the four primary functions listed below:

1. In-Phase Power Combining
2. Head / Body Switching
3. Patient Safety Power Monitoring
4. PIN Switch Bias Filtering

The insertion loss contributed by the Output Conditioning Module in a 1.0T RFI should not exceed 0.35dB while the insertion loss contributed by the Output Conditioning Module in a 1.5T RFI should not exceed 0.25dB.

5-1 In-Phase Power Combining

The outputs of two power amplifiers are combined by way of a 0° power combiner (T1). The inputs to the power combiner are AMP1 IN (J1) and AMP2 IN (J2). Ideally, the AMP1 IN and AMP2 IN signals should be equal in phase and amplitude. T1 functions in such a way to sum currents from the two inputs, hence, the output impedance is 25 ohms and the output voltage is equal to the input voltage.

The output is matched to 50 ohms by way of an “L” network. L1 is an inductive transmission line that is implemented with a brass rod that is mounted on standoffs to the bottom of the board. This line has distributed capacitance to ground and its location relative to the board and the sheet metal chassis is critical. The length of all the standoffs, including the 5/8” standoffs on the PCB to chassis mounting holes, are critical. Also, the proper orientation of the rod L1 is critical. L1 should be installed with the solder lugs of the ring tongue terminals facing away from the board so that the spacing between the brass rod and the board is maximized.

Any amplitude or phase difference between the AMP1 IN and AMP2 IN signals shows up as a differential voltage across the dump resistor R34. At any instant, the voltage across the dump resistor is equal to half the difference between the two input voltages. For Continuous Wave (CW), the average power dissipated in the dump resistor can be expressed as the instantaneous power dissipation averaged over one period.

$$P_D = \frac{\omega}{2\pi} \int_0^{\frac{2\pi}{\omega}} \frac{V_D^2(t)}{R_D} dt \quad \text{Eq. 1}$$

where,

$$R_D = 25\Omega ,$$

$$V_D = \frac{|V_1(t) - V_2(t)|}{2}$$

For pulsed power, Equation 1 represents Peak Envelope Power (PEP).

5-1 In-Phase Power Combining (continued)

The power dissipated by the dump resistor and the resulting heat measured by the thermistor mounted on the dump resistor is used to obtain the dump resistor overtemp signal. The processor module monitors a scaled representation of the thermistor voltage and responds with a fault in the event that the voltage through the thermistor goes out of limits. This would indicate that excessive power was being dissipated in the dump resistor. The dump resistor, R_D , has a value of 250 ohms. The dump power fault threshold is set at 750W PEP. This corresponds to 75W average power, assuming a 10% duty cycle.

5-2 Head / Body Switching

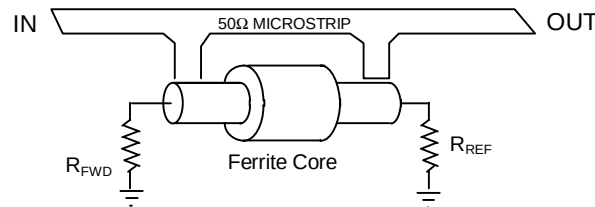
Switching between head and body mode is accomplished by way of 2 vacuum relays. The relay driver circuitry is located on the Input Conditioning Board. The output of the power combiner is connected to the normally open pin of both relays. The normally closed pin of each relay is grounded. For head mode, the head relay is energized and the body relay is off. Contact is made between the output of the power combiner and the common pin (pole) of the head relay. For body mode, the body relay is energized and the head relay is off. Contact is made between the output of the power combiner and the common pin (pole) of the body relay. In each case, the common pin (pole) of the off relay is grounded. This arrangement provides better head/body isolation than would be possible with a single relay of this type.

5-3 Patient Safety Power Monitoring

Forward power monitoring is accomplished by way of two pairs of directional couplers (a pair for the head mode signal path and a pair for the body mode signal path). Each pair of directional couplers provides 2 independent forward power samples for redundant patient safety monitoring. The 1.0T head directional couplers, DC1 & 2, have a coupling coefficient of -35dB (+/- 0.5dB) while the 1.5T head directional couplers, DC1 & 2, have a coupling coefficient of -40dB (+/- 0.5dB). The 1.0T body directional couplers, DC 3 &4, have a coupling coefficient of -46dB (+/- 0.5dB) while the 1.5T body directional couplers, DC 3 &4, have a coupling coefficient of -50dB (+/- 0.5dB).

The head and body couplers both work on the same basic principals. The shield of a length of coax is connected in parallel with a length of the primary transmission line (in this case a 50 ohm microstrip). During operation, a voltage will exist across the length of the transmission line (and hence across the length of the shield) due to current flow and phase delay. A ferrite core suppresses common mode currents in the coax causing the coax to function like a transformer.

5-3 Patient Safety Power Monitoring (continued)



CURRENT FLOW
ILLUSTRATION 5-1

The voltage across the length of the shield causes a small current to flow along the shield that in turn causes a magnetically coupled current to flow through the center conductor. This current flows from the reverse termination (R_{REF}) to the forward termination (R_{FWD}) and results in voltages of opposite polarities at the terminations. See Illustration 5-1. Voltage is electrically coupled through the distributed capacitance of the coax with the same polarity at the forward and reverse terminations. In the case of zero reflected power, the voltages due to magnetic and capacitive coupling add at the forward port and cancel at the reflected port. The forward termination consists of a variable resistive attenuator pad that is adjusted in production to obtain precisely the correct coupling coefficient.

5-4 PIN Switch Bias Filtering

During transmission, a positive DC bias is applied to the output in order to forward bias a PIN diode T/R switch. While receiving, a negative DC voltage is applied to the output in order to reverse bias the PIN diode T/R switch. In head mode, the DC bias current is fed through the HEAD TR port (J9) and passes through L4 and L2 on its way to the HEAD OUTPUT (J3). The DC is blocked by C3 and C11. C6 and C7 interact with L2 and L4 to provide enough filtering to prevent the PIN switch driver from being infiltrated by RF. In body mode, the DC bias current is fed through the BODY TR port (J10) and passes through L5 and L3 on its way to the BODY OUTPUT (J4). The DC is blocked by C4 and C5. C8 and C9 interact with L3 and L5 to provide enough filtering to prevent the PIN switch driver from being infiltrated by RF.

REVISION HISTORY

REV	DATE	AUTHOR	PRIMARY REASONS FOR CHANGE
A	Jan. 27, 1998	Erbtec (KK.)	Preliminary release.
0	June 30, 1998	Erbtec Engineering	Initial Release
1	Feb. 9, 1999	Resa Lambert	Added information to Section 1- Overview, which further defines main function of the RFI and note stating the envelope feedback is no longer used through the SSM.
2	Oct 13, 1999	M. Keber	Added correct proprietary heading to document.
3	March 22, 2000	Resa Lambert	Added drawing and technical verbiage to Section 1.
4	May 9, 2001	Don Thome	Added information throughout document.
5	January 4, 2002	Don Thome	Corrected error in Section 2-3-4.
6	January 17, 2002	Don Thome	Corrected errors in Sections 2-1, 2-4 and 5.
7	May 13, 2002	Don Thome	Corrected drawing error in Illustration 1-2.